MODULE 3: ASSET BASED ACCOUNTING STANDARDS II

AS 16 – BORROWING COST

Illustration 1

PRM Ltd. obtained a loan from a bank for Rs.120 lakhs on 30-04-20X1. It was utilised as follows:

Particulars	***Amount (Rs.in lakhs)***
Construction of a shed	*50*
Purchase of a machinery	*40*
Working Capital	*20*
Advance for purchase of truck	*10*

Construction of shed was completed in March 20X2. The machinery was installed on the date of acquisition. Delivery of truck was not received. Total interest charged by the bank for the year ending 31-03-20X2 was Rs.18 lakhs. Show the treatment of interest.

Solution

Qualifying Asset as per AS 16 = Rs.50 lakhs (construction of a shed)

Borrowing cost to be capitalised = 18 x 50/120 = Rs.7.5 lakhs

Interest to be debited to Profit or Loss account = Rs.(18 – 7.5) lakhs = Rs.10.5 lakhs

Illustration 2

X Ltd. began construction of a new building on 1st January, 20X1. It obtained Rs.1 lakh special loan to finance the construction of the building on 1st January, 20X1 at an interest rate of 10%. The company's other outstanding two non-specific loans were:

Amount	***Rate of Interest***
Rs.5,00,000	*11%*
Rs.9,00,000	*13%*

The expenditures that were made on the building project were as follows:

		Rs.
January	*20X1*	*2,00,000*
April	*20X1*	*2,50,000*
July	*20X1*	*4,50,000*
December	*20X1*	*1,20,000*

Building was completed by 31st December 20X1. *Following the principles prescribed in AS 16 'Borrowing Cost,' calculate the amount of interest to be capitalised and pass one Journal Entry for capitalising the cost and borrowing cost in respect of the building.*

Solution

(i) Computation of weighted average accumulated expenses

		Rs.
Rs.2,00,000 x 12 / 12	=	2,00,000
Rs.2,50,000 x 9 / 12	=	1,87,500
Rs.4,50,000 x 6 / 12	=	2,25,000
Rs.1,20,000 x 1 / 12	=	10,000
		6,22,500

(ii) Calculation of weighted average interest rate other than for specific borrowings

Amount of loan (Rs.)	**Rate of interest**		**Amount of interest (Rs.)**
5,00,000	11%	=	55,000
9,00,000	13%	=	1,17,000
14,00,000			1,72,000
Weighted average rate of interest $\left(\frac{1,72,000}{14,00,000} \times 100\right)$		=	12.285% (approx.)

(iii) Interest on weighted average accumulated expenses

		Rs.
Specific borrowings (Rs.1,00,000 x 10%)	=	10,000
Non-specific borrowings (Rs.5,22,500* x 12.285%)	=	64,189
Amount of interest to be capitalised	=	74,189

(iv) Total expenses to be capitalized for building

	Rs.
Cost of building Rs. (2,00,000 + 2,50,000 + 4,50,000 + 1,20,000)	10,20,000
Add: Amount of interest to be capitalised	74,189
	10,94,189

(v) Journal Entry

Date	Particulars		Dr. (Rs.)	Cr. (Rs.)
31.12. 20X1	Building account	Dr.	10,94,189	
	To Bank account			10,94,189
	(Being amount of cost of building and borrowing cost thereon capitalised)			

Illustration 3

The company has obtained Institutional Term Loan of Rs.580 lakhs for modernisation and renovation of its Plant & Machinery. Plant & Machinery acquired under the modernisation scheme and installation completed on 31st March, 20X2 amounted to Rs.406 lakhs, Rs.58 lakhs has been advanced to suppliers for additional assets and the balance loan of Rs.116 lakhs has been utilised for working capital purpose. The Accountant is on a dilemma as to how to account for the total interest of Rs.52.20 lakhs incurred during 20X1-20X2 on the entire Institutional Term Loan of Rs.580 lakhs.

Solution

As per para 6 of AS 16 'Borrowing Costs', borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalised as part of the cost of that asset. Other borrowing costs should be recognised as an expense in the period in which they are incurred.

A qualifying asset is an asset that necessary takes a substantial period of time* to get ready for its intended use or sale.

The treatment for total interest amount of Rs.52.20 lakhs can be given as:

Purpose	**Nature**	**Interest to be capitalised**	**Interest to be charged to profit and loss account**
		Rs.in lakhs	**Rs.in lakhs**
Modernisation and renovation of plant and machinery	Qualifying asset	$**52.20 \times \frac{406}{580} = 36.54$	
		$**52.20 \times \frac{[illegible]}{580} = 5.22$	
Advance to supplies for additional assets	Qualifying asset		$52.20 \times \frac{116}{580} = 10.44$
Working Capital	Not a qualifying asset		
		41.76	10.44

** A substantial period of time primarily depends on the facts and circumstances of each case. However, ordinarily, a period of twelve months is considered as substantial period of time unless a shorter or longer period can be justified on the basis of the facts and circumstances of the case.*

*** It is assumed in the above solution that the modernisation and renovation of plant and machinery will take substantial period of time (i.e. more than twelve months). Regarding purchase of additional assets, the nature of additional assets has also been considered as qualifying assets. Alternatively, the plant and*

machinery and additional assets may be assumed to be non-qualifying assets on the basis that the renovation and installation of additional assets will not take substantial period of time. In that case, the entire amount of interest, Rs.52.20 lakhs will be recognised as expense in the profit and loss account for year ended 31st March, 20X2.

Illustration 4

Take Ltd. has borrowed Rs.30 lakhs from State Bank of India during the financial year 20X1-20X2. The borrowings are used to invest in shares of Give Ltd., a subsidiary company of Take Ltd., which is implementing a new project, estimated to cost Rs.50 lakhs. As on 31st March, 20X2, since the said project was not complete, the directors of Take Ltd. resolved to capitalise the interest accruing on borrowings amounting to Rs.4 lakhs and add it to the cost of investments. Comment.

Solution

As per AS 13 (Revised) "Accounting for Investments", the cost of investment includes acquisition charges such as brokerage, fees and duties. In the present case, Take Ltd. has used borrowed funds for purchasing shares of its subsidiary company Give Ltd. Rs.4 lakhs interest payable by Take Ltd. to State Bank of India cannot be called as acquisition charges, therefore, cannot be constituted as cost of investment.

Further, as per para 3 of AS 16 "Borrowing Costs", a qualifying asset is an asset that necessarily takes a substantial period of time to get ready for its intended use or sale. Since, shares are ready for its intended use at the time of sale, it cannot be considered as qualifying asset that can enable a company to add the borrowing cost to investments. Therefore, the directors of Take Ltd. cannot capitalise the borrowing cost as part of cost of investment. Rather, it has to be charged to the Statement of Profit and Loss for the year ended 31st March, 20X2.

Reference: The students are advised to refer the full text of AS 16 "Borrowing Costs" (issued 2000).

5. *On 1st April, 20X1, Amazing Construction Ltd. obtained a loan of Rs.32 crores to be utilised as under:*

(i)	*Construction of sealink across two cities:*		
	(work was held up totally for a month during the year due to high water levels)	:	*Rs.25*
(ii)	*Purchase of equipments and machineries*	:	*Rs.3 crores*
(iii)	*Working capital*	:	*Rs.2 crores*
(iv)	*Purchase of vehicles*	:	*Rs.50,00,000*
(v)	*Advance for tools/cranes etc.*	:	
(vi)	*Purchase of technical know-how*	:	*Rs.50,00,000*
(vii)	*Total interest charged by the bank for the year ending 31st March, 20X2*	:	*Rs.1 crores*

Show the treatment of interest by Amazing Construction Ltd.

Solution

According to AS 16 'Borrowing costs', qualifying asset is an asset that necessarily takes substantial period of time to get ready for its intended use.

Borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalised as part of the cost of that asset. Other borrowing costs should be recognised as an expense in the period in which they are incurred.

The treatment of interest by Amazing Construction Ltd. can be shown as:

	Qualifying Asset	**Interest to be capitalised Rs.**	**Interest to be charged to Profit & Loss A/c Rs.**	
Construction of sea-link	Yes	62,50,000		[80,00,000x(25/32)]
Purchase of	No		7,50,000	[80,00,000x(3/32)]

equipment and machineries				
Working capital	No		5,00,000	[80,00,000x(2/32)]
Purchase of vehicles	No		1,25,000	[80,00,000x(0.5/32)]
Advance for tools, cranes etc.	No		1,25,000	[80,00,000x(0.5/32)]
Purchase of technical know-how	No		2,50,000	[80,00,000x(1/32)]
Total		62,50,000	17,50,000	

*It is assumed that work held up for a month due to high water level is normal during the construction of sealink and capitalization of borrowing cost should not be suspended for necessary temporary delay.

6. *Rainbow Limited borrowed an amount of Rs.150 crores on 1.4.20X1 for construction of boiler plant @ 11% p.a. The plant is expected to be completed in 4 years. Since the weighted average cost of capital is 13% p.a., the accountant of Rainbow Ltd. capitalized Rs.19.50 crores for the accounting period ending on 31.3.20X2. Due to surplus fund out of Rs.150 crores, income of Rs.3.50 crores were earned and credited to profit and loss account. Comment on the above treatment of accountant with reference to relevant accounting standard.*

Solution

Para 10 of AS 16 'Borrowing Costs' states "To the extent that funds are borrowed specifically for the purpose of obtaining a qualifying asset, the amount of borrowing costs eligible for capitalization on that asset should be determined as the actual borrowing costs incurred on that borrowing during the period less any income on the temporary investment of those borrowings."

The capitalization rate should be the weighted average of the borrowing costs applicable to the borrowings of the enterprise that are outstanding during the period, other than borrowings made specifically for the purpose of obtaining a qualifying asset.

Thus, the treatment of accountant of Rainbow Ltd. is incorrect.

Amount of borrowing costs capitalized should be calculated as follows:

Particulars	**Rs.in crores**
Actual interest for 20X1-20X2 (11% of Rs.150 crores)	16.50
Less: Income on temporary investment from specific borrowings	(3.50)
Borrowing costs to be capitalized during year 20X1-20X2	13.00

7. *Harish Construction Company is constructing a huge building project consisting of four phases. It is expected that the full building will be constructed over several years but Phase I and Phase II of the building will be started as soon as they are completed.*

Following is the detail of the work done on different phases of the building during the current year:

(Rs.in lakhs)

	Phase I	***Phase II***	***Phase III***	***Phase IV***
	Rs.	***Rs.***	***Rs.***	***Rs.***
Cash expenditure	*10*	*30*	*25*	*30*
Building purchased	*24*	*34*	*30*	*38*
Total expenditure	*34*	*64*	*55*	*68*
Total expenditure of all phases				*221*
Loan taken @ 15% at the beginning of the year				*200*

During mid of the current year, Phase I and Phase II have become operational.

Find out the total amount to be capitalized and to be expensed during the year.

Solution

Computation of amount to be capitalized

No.	Particulars	Rs.
1.	Interest expense on loan Rs.2,00,00,000 at 15%	30,00,000
2.	Total cost of Phases I and II (Rs.34,00,000 +64,00,000)	98,00,000
3.	Total cost of Phases III and IV (Rs. 55,00,000 + Rs.68,00,000)	1,23,00,000
4.	Total cost of all 4 phases	2,21,00,000
5.	Total loan	2,00,00,000
6.	Interest on loan used for Phases I & II, based on proportionate Loan amount = $\frac{30,00,000}{2,21,00,000} \times 98,00,000$	13,30,317 (approx.)
7.	Interest on loan used for Phases III & IV, based on proportionate Loan amount = $\frac{30,00,000}{2,21,00,000} \times 1,23,00,000$	16,69,683 (approx.)

Accounting treatment

For Phase I and Phase II

Since Phase I and Phase II have become operational at the mid of the year, half of the interest amount of Rs.6,65,158.50 (i.e. Rs.13,30,317/2) relating to Phase I and Phase II should be capitalized (in the ratio of asset costs 34:64) and added to respective assets in Phase I and Phase II and remaining half of the interest amount of Rs.6,65,158.50 (i.e. Rs.13,30,317/2) relating to Phase I and Phase II should be expensed during the year.

For Phase III and Phase IV

Interest of Rs.16,69,683 relating to Phase III and Phase IV should be held in Capital Work-in-Progress till assets construction work is completed, and thereafter capitalized in the ratio of cost of assets. No part of this interest amount should be charged/expensed off during the year since the work on these phases has not been completed yet.

8. Expert Limited issued 12% secured debentures of Rs.100 lakhs on 01.06.20X1.

Money raised from debentures to be utilized as under:

Intended Purpose	Amount Rs.in lakhs
Construction of factory building	40
Working Capital	30
Purchase of Machinery	15
Purchase of Furniture	2
Purchase of truck	13

Additional Information:

(i) Interest on debentures for the Financial Year 20X1-20X2 was paid by the Company.

(ii) During the year, the company invested idle fund of Rs.5 lakhs (out of the money raised from debentures) in Bank's fixed deposit and earned interest of Rs.50,000.

(iii) In March, 20X2 construction of factory building was not completed (it is expected that it will take another 6 months).

(iv) In March 20X2, Machinery was installed and ready for its intended use.

(v) Furniture was put to use at the end of March 20X2.

(vi) Truck is going to be received in April, 20X2.

You are required to show the treatment of interest as per AS 16 in respect of borrowing cost for the year ended 31st March, 20X2 in the Books of Expert Limited.

Solution

According to AS 16 "Borrowing Costs", a qualifying asset is an asset that necessarily takes a substantial period of time to get ready for its intended use. As per the Standard, borrowing costs that are directly attributable to the acquisition, construction or production of a qualifying asset should be capitalized as part of the cost of that asset. The amount of borrowing costs eligible for capitalization should be determined in accordance with this Standard. Other borrowing costs should be recognized as an expense in the period in which they are incurred. It also states that to the extent that funds are borrowed specifically for the purpose of obtaining a qualifying asset, the amount of borrowing costs eligible for capitalization on that asset should

be determined as the actual borrowing costs incurred on that borrowing during the period less any income on the temporary investment of those borrowings.

Thus, eligible borrowing cost = Rs.10,00,000 (100 lakhs x 12% x 10/12) – Rs.50,000 = Rs.9,50,000

Particulars	**Nature of assets**	**Interest to be capitalized (Rs.)**	**Interest to be charged to Profit & Loss Account (Rs.)**
Construction of factory building	Qualifying Asset	9,50,000x40/100 = Rs.3,80,000	NIL
Purchase of Machinery	Not a Qualifying Asset	NIL	9,50,000x15/100 = 1,42,500
Purchase of and furniture	Not a Qualifying Asset	NIL	9,50,000x2/100 =19,000
Purchase of truck	Not a Qualifying Asset	NIL	9,50,000x13/100 = 1,23,500
Working Capital	Not a Qualifying Asset	NIL	9,50,000x30/100 = Rs.2,85,000
Total		Rs.3,80,000	Rs.5,70,000

Example 1

Annual lease rents = *Rs. 50,000 at the end of each year.*

Lease period = *5 years;*

Guaranteed residual value = *Rs. 25,000*

Unguaranteed residual value (UGR) = *Rs.15,000*

Fair Value at the inception (beginning) of lease = Rs. 2,00,000 Interest rate implicit on lease is computed below:

Interest rate implicit on lease is a discounting rate at which present value of minimum lease payments and unguaranteed residual value is Rs. 2 lakhs.

PV of minimum lease payments and unguaranteed residual value at guessed rate 10%

Year	MLP + UGR Rs.	DF (10%)	PV Rs.
1	50,000	0.909	45,450
2	50,000	0.826	41,300
3	50,000	0.751	37,550
4	50,000	0.683	34,150
5	50,000	0.621	31,050
5	25,000	0.621	15,525
5	15,000	0.621	9,315
			2,14,340

PV of minimum lease payments and unguaranteed residual value at guessed rate 14%

Year	MLP + UGR Rs.	DF (14%)	PV Rs.
1	50,000	0.877	43,850
2	50,000	0.769	38,450
3	50,000	0.675	33,750
4	50,000	0.592	29,600
5	50,000	0.519	25,950
5	25,000	0.519	12,975
5	15,000	0.519	7,785
			1,92,360

Interest rate implicit on lease is computed below by interpolation:

$$\text{Interest rate implicit on lease} = 10\% + \frac{14\% - 10\%}{2,14,340 - 1,92,360} \times (2,14,340 - 2,00,000) = 12.6\%$$

Example 2

Annual lease rents = Rs. 50,000 at the end of each year.

Lease period = 5 years;

Guaranteed residual value = Rs. 25,000 Unguaranteed residual value

(UGR) = Rs. 15,000

Fair Value at the inception (beginning) of lease = Rs. 2,00,000

Interest rate implicit on lease is =12.6%

Present value of minimum lease payment is computed below:

Year	MLP Rs.	DF (12.6%)	PV Rs.
1	50,000	0.890	44,500
2	50,000	0.790	39,500
3	50,000	0.700	35,000

4	*50,000*	*0.622*	*31,100*
5	*50,000*	*0.552*	*27,600*
5	*25,000*	*0.552*	*13,800*
			1,91,500

Present value of minimum lease payment = Rs. 1,91,500

Fair value of leased asset = Rs. 2,00,000

The accounting entry at the inception of lease to record the asset taken on finance lease in books of lessee is suggested below:

	Rs.	*Rs.*
Asset A/c Dr.	*1,91,500*	
To Lessor (Lease Liability) A/c		*1,91,500*
(Being recognition of finance lease as asset and liability)		

Example 3

Using data for example 2 and assuming zero residual value, allocation of finance charge over lease period is shown below:

Year	***Minimum Lease Payments*** ***Rs.***	***Finance Charge (12.6%)*** ***Rs.***	***Principal*** ***Rs.***	***Principal due*** ***Rs.***
0	*--*	*--*	*--*	*1,91,500*
1	*50,000*	*24,129*	*25,871*	*1,65,629*
2	*50,000*	*20,869*	*29,131*	*1,36,498*
3	*50,000*	*17,199*	*32,801*	*1,03,697*
4	*50,000*	*13,066*	*36,934*	*66,763*
5	*75,000*	*8,237*[*]	*66,763*	
	2,75,000	*83,500*	*1,91,500*	

[*] The difference between this figure and finance charge [66,763×12.6%=8412] is due to approximation in computation.

Accounting entries in year 1 to recognise the finance charge in books of lessee are suggested below:

		Rs.	Rs.
Finance Charge A/c	*Dr.*	*24,129*	
To Lessor			*24,129*
(Being finance charge due for the year)			
Lessor	*Dr.*	*50,000*	
To Bank A/c			*50,000*
(Being payment of lease rent for the year)			
P & L A/c	*Dr.*	*24,129*	
To Finance Charge A/c			*24,129*
(Being recognition of finance charge as expense for the year)			

Example 4

In example 2, suppose unguaranteed residual value is not determinable and lessee's incremental borrowing rate is 10%.

Since interest rate implicit on lease is discounting rate at which present value of minimum lease payment and present value of unguaranteed residual value equals the fair value, interest rate implicit on lease cannot be determined unless unguaranteed residual value is known. If interest rate implicit on lease is not determinable, the present value of minimum lease payments should be determined using lessee's incremental borrowing rate.

Present value of minimum lease payment using lessee's incremental borrowing rate 10% is computed below:

Year	***MLP Rs.***	***DF (10%)***	***PV Rs.***
1	*50,000*	*0.909*	*45,450*

2	50,000	0.826	41,300
3	50,000	0.751	37,550
4	50,000	0.683	34,150
5	50,000	0.621	31,050
5	25,000	0.621	15,525
			2,05,025

Present value of minimum lease payment = Rs. 2,05,025

Fair value of leased asset = Rs. 2,00,000

The accounting entry at the inception of lease to record the asset taken on finance lease in books of lessee is suggested below:

	Rs.	**Rs.**
Asset A/c Dr.	*2,00,000*	
To Lessor (Lease Liability)		*2,00,000*
(Being recognition of finance lease as asset and liability)		

Since the liability is recognised at fair value Rs. 2 lakh (total principal), we need to ascertain a discounting rate at which present value minimum lease payments equals Rs. 2 lakh. The discounting rate can then be used for allocation of finance charge over lease period.

PV of minimum lease payments at guessed rate 12%.

Year	**Minimum Lease Payments Rs.**	**DF (12%)**	**PV Rs.**
1	50,000	0.893	44,650
2	50,000	0.797	39,850
3	50,000	0.712	35,600
4	50,000	0.636	31,800
5	50,000	0.567	28,350

5	25,000	0.567	14,175
			1,94,425

Required discounting rate = $10\% + \frac{12\% - 10\%}{2,05,025 - 1,94,425} \times (2,05,025 - 2,00,000) = 10.95\%$

Allocation of finance charge over lease period is shown below:

Year	Minimum Lease Payments Rs.	Finance Charge (10.95%) Rs.	Principal Rs.	Principal due Rs.
0	--	--	--	2,00,000
1	50,000	21,900	28,100	1,71,900
2	50,000	18,823	31,177	1,40,723
3	50,000	15,409	34,591	1,06,132
4	50,000	11,621	38,379	67,753
5	75,000	7,247*	67,753	
	2,75,000	75,000	2,00,000	

Accounting entries in year 1 to recognise the finance charge in books of lessee are suggested below:

		Rs.	Rs.
Finance Charge A/c	Dr.	21,900	
To Lessor			21,900
(Being finance charge due for the year)			
Lessor	Dr.	50,000	
To Bank A/c			50,000
(Being payment of lease rent for the year)			

* The difference between this figure & finance charge [67,753×10.95% = 7418] is due to approximation in computation

P & L A/c	*Dr.*	*21,900*	
To Finance Charge			*21,900*
(Being recognition of finance charge as expense for the year)			

Illustration 1

S. Square Private Limited has taken machinery on finance lease from S.K. Ltd. The information is as under:

Lease term = 4 years

Fair value at inception of lease = Rs. 20,00,000

Lease rent = Rs. 6,25,000 p.a. at the end of year

Guaranteed residual value = Rs. 1,25,000

Expected residual value = Rs. 3,75,000

Implicit interest rate = 15%

Discounted rates for 1st year, 2nd year, 3rd year and 4th year are 0.8696, 0.7561, 0.6575 and 0.5718 respectively.

Calculate the value of the lease liability as per AS-19 and disclose impact of this on Balance sheet and Profit & loss account at the end of year 1

Solution

According to para 11 of AS 19 "Leases", the lessee should recognise the lease as an asset and a liability at an amount equal to the lower of the fair value of the leased asset at the inception of the finance lease and the present value of the minimum lease payments from the standpoint of the lessee.

In calculating the present value of the minimum lease payments the discount rate is the interest rate implicit in the lease. Present value of minimum lease payments will be calculated as follows:

Year	Minimum Lease Payment Rs.	Implicit interest rate (Discount rate @15%)	Present value Rs.
1	6,25,000	0.8696	5,43,500
2	6,25,000	0.7561	4,72,563
3	6,25,000	0.6575	4,10,937
4	7,50,000*	0.5718	4,28,850
Total	26,25,000		18,55,850

Present value of minimum lease payments *Rs.* 18,55,850 is less than fair value at the inception of lease i.e. Rs. 20,00,000, therefore, the asset and corresponding lease liability should be recognised at *Rs.* 18,55,850 as per AS 19.

Illustration 2

Prakash Limited leased a machine to Badal Limited on the following terms:

		(Rs. In lakhs)
(i)	*Fair value of the machine*	*28.3*
(ii)	*Lease term*	*5 years*
(iii)	*Lease rental per annum*	*8.00*
(iv)	*Guaranteed residual value*	*1.60*
(v)	*Expected residual value*	*3.00*
(vi)	*Internal rate of return*	*15%*

Discounted rates for 1st year to 5th year are 0.8696, 0.7561, 0.6575, 0.5718, and 0.4972 respectively.

Ascertain Unearned Finance Income.

Solution

As per AS 19 on Leases, ***unearned finance income*** is the difference between (a) the ***gross investment*** in the lease and (b) the present value of minimum lease payments under a finance lease from the standpoint of the lessor; and any unguaranteed residual value accruing to the lessor, at the interest rate implicit in the lease.

Where:

(a) ***Gross investment*** in the lease is the aggregate of (i) minimum lease payments from the stand point of the lessor and (ii) any unguaranteed residual value accruing to the lessor.

Gross investment = Minimum lease payments + Unguaranteed residual value

= [Total lease rent + Guaranteed residual value (GRV)] + Unguaranteed residual value (URV)

= [(Rs. 8,00,000 × 5 years) + Rs. 1,60,000] + Rs. 1,40,000

= Rs. 43,00,000 (a)

(b) Table showing present value of (i) Minimum lease payments (MLP) and (ii) Unguaranteed residual value (URV).

Year	MLP inclusive of URV (Rs.)	Internal rate of return (Discount factor @ 15%)	Present Value (Rs.)
1	8,00,000	0.8696	6,95,680
2	8,00,000	0.7561	6,04,880
3	8,00,000	0.6575	5,26,000
4	8,00,000	0.5718	4,57,440
5	8,00,000	0.4972	3,97,760
	1,60,000 (GRV)	0.4972	79,552
	41,60,000		27,61,312 (i)
	1,40,000 (URV)	0.4972	69,608 (ii)
	43,00,000	(i)+ (ii)	28,30,920 (b)

Unearned Finance Income (a) - (b) = Rs. 43,00,000 – Rs. 28,30,920= Rs. 14,69,080.

Illustration 3

A Ltd. sold machinery having WDV of Rs. 40 lakhs to B Ltd. for Rs. 50 lakhs and the same machinery was leased back by B Ltd. to A Ltd. The lease back is operating lease. Comment if –

(a) Sale price of Rs. 50 lakhs is equal to fair value.

(b) Fair value is Rs. 60 lakhs.

(c) Fair value is Rs. 45 lakhs and sale price is Rs. 38 lakhs.

(d) Fair value is Rs. 40 lakhs and sale price is Rs. 50 lakhs.

(e) Fair value is Rs. 46 lakhs and sale price is Rs. 50 lakhs

(f) Fair value is Rs. 35 lakhs and sale price is Rs. 39 lakhs.

Solution

Following will be the treatment in the given cases:

(a) When sales price of Rs. 50 lakhs is equal to fair value, A Ltd. should immediately recognise the profit of Rs. 10 lakhs (i.e. 50 – 40) in its books.

(b) When fair value is Rs. 60 lakhs then also profit of Rs. 10 lakhs should be immediately recognised by A Ltd.

(c) When fair value of leased machinery is Rs. 45 lakhs & sales price is Rs. 38 lakhs, then loss of Rs. 2 lakhs (40 – 38) to be immediately recognised by A Ltd. in its books provided loss is not compensated by future lease payment, otherwise defer and amortise the loss.

(d) When fair value is Rs. 40 lakhs & sales price is Rs. 50 lakhs then, profit of Rs. 10 lakhs is to be deferred and amortised over the lease period.

(e) When fair value is Rs. 46 lakhs & sales price is Rs. 50 lakhs, profit of Rs. 6 lakhs (46 - 40) to be immediately recognised in its books and balance profit of Rs. 4 lakhs (50-46) is to be amortised/deferred over lease period.

(f) When fair value is Rs. 35 lakhs & sales price is Rs. 39 lakhs, then the loss of Rs. 5 lakhs (40-35) to be immediately recognised by A Ltd. in its books and profit of Rs. 4 lakhs (39-35) should be amortised/deferred over lease period.

4. Classify the following into either operating or finance lease:

(i) Lessee has option to purchase the asset at lower than fair value, at the end of lease term;

(ii) Economic life of the asset is 7 years, lease term is 6 years, but asset is not acquired at the end of the lease term;

(iii) Economic life of the asset is 6 years, lease term is 2 years, but the asset

is of special nature and has been procured only for use of the lessee;

(iv) Present value (PV) of Minimum lease payment (MLP) = "X". Fair value of the asset is "Y".

Solution:

(i) If it becomes certain at the inception of lease itself that the option will be exercised by the lessee, it is a Finance Lease.

(ii) The lease will be classified as a finance lease, since a substantial portion of the life of the asset is covered by the lease term.

(iii) Since the asset is procured only for the use of lessee, it is a finance lease.

(iv) The lease is a finance lease if X = Y, or where X substantially equals Y.

5. *A machine was given on 3 years operating lease by a dealer of the machine for equal annual lease rentals to yield 30% profit margin on cost Rs. 1,50,000. Economic life of the machine is 5 years and output from the machine are estimated as 40,000 units, 50,000 units, 60,000 units, 80,000 units and 70,000 units consecutively for 5 years. Straight line depreciation in proportion of output is considered appropriate. Compute the following:*

(i) Annual Lease Rent

(ii) Lease Rent income to be recognized in each operating year and

(iii) Depreciation for 3 years of lease.

Solution:

(i) Annual lease rent

Total lease rent

$$= 130\% \text{ of Rs. } 1,50,000 \times \frac{\text{Output during lease period}}{\text{Total output}}$$

= 130% of Rs. 1,50,000 x (40,000 +50,000+ 60,000)/(40,000 + 50,000 + 60,000 + 80,000 + 70,000)

= 1,95,000 x 1,50,000 units/3,00,000 units = Rs. 97,500

Annual lease rent = Rs. 97,500 / 3 = Rs. 32,500

(ii) Lease rent Income to be recognized in each operating year

Total lease rent should be recognised as income in proportion of output during lease period, i.e. in the proportion of 40 : 50 : 60.

Hence income recognised in years 1, 2 and 3 will be as:

Year 1 Rs. 26,000,

Year 2 Rs. 32,500 and

Year 3 Rs. 39,000.

(iii) Depreciation for three years of lease

Since depreciation in proportion of output is considered appropriate, the depreciable amount Rs. 1,50,000 should be allocated over useful life 5 years in proportion of output, i.e. in proportion of 40 : 50 : 60 : 80 :
70 .

Depreciation for year 1 is Rs. 20,000, year 2 = 25,000 and year 3 = 30,000**.**

6. *Lessee Ltd. took a machine on lease from Lessor Ltd., the fair value being Rs. 7,00,000.*

The economic life of machine as well as the lease term is 3 years. At the end of each year Lessee Ltd. pays Rs. 3,00,000. The Lessee has guaranteed a residual value of Rs. 22,000 on expiry of the lease to the Lessor. However, Lessor Ltd., estimates that the residual value of the machinery will be only Rs. 15,000. The implicit rate of return is 15% p.a. and present value factors at 15% are 0.869, 0.756 and 0.657 at the end of first, second and third years respectively.

Calculate the value of machinery to be considered by Lessee Ltd. and the finance charges in each year.

Solution:

As per para 11 of AS 19 "Leases", the lessee should recognize the lease as an asset and a liability at the inception of a finance lease. Such recognition should be at an amount equal to the fair value of the leased asset at the inception of lease. However, if the fair value of the leased asset exceeds the present value of minimum lease payment from the standpoint of the lessee, the amount recorded as an asset and liability should be the present value of

minimum lease payments from the standpoint of the lessee.

Computation of Value of machinery:

Present value of minimum lease payment = Rs. 6,99,054 (See working note below)

Fair value of leased asset = Rs. 7,00,000

Therefore, the recognition will be at the lower of the two i.e. 6,99,054

Working Note - Present value of minimum lease payments:

Annual lease rental × PVIF+ Present value of guaranteed residual value

= Rs. 3,00,000 × (0.869 + 0.756 + 0.657) + Rs. 22,000 × 0.657

= Rs. 6,84,600 + Rs. 14,454 = 6,99,054

Computation of finance charges:

Year	Finance charge	Payment	Reduction in outstanding liability	Outstanding liability
1st Year beginning	–	–	–	6,99,054
End of 1st year	1,04,858	3,00,000	1,95,142	5,03,912
End of 2nd year	75,587	3,00,000	2,24,413	2,79,499
End of 3rd year	41,925	3,00,000	2,58,075	21,424

7. *B&P Ltd. availed a lease from N&L Ltd. The conditions of the lease terms are as under:*

   (i) Lease period is 3 years, in the beginning of the year 2009, for equipment costing Rs. 10,00,000 and has an expected useful life of 5 years.

   (ii) The Fair market value is also Rs. 10,00,000

   (iii) The property reverts back to the lessor on termination of the lease.

   (iv) The unguaranteed residual value is estimated at Rs. 1,00,000 at the end of the year 2011.

(v) 3 equal annual payments are made at the end of each year.

(vi) Consider IRR = 10%.

The present value off Rs. 1 due at the end of 3rd year at 10% rate of interest is Rs. 0.7513. The present value of annuity of Rs. 1 due at the end of 3rd year at 10% IRR is Rs. 2.4868.

State whether the lease constitute finance lease and also calculate unearned finance income.

Solution:

Computation of annual lease payment:

Particulars	**Rs.**
Cost of equipment	10,00,000
Unguaranteed residual value	1,00,000
Present value of unguaranteed residual value (Rs. 1,00,000 x 0.7513)	75,130
Present value of lease payments (Rs. 10,00,000 - Rs. 75,130)	9,24,870
Present value of annuity for three years is 2.4868	
Annual lease payment [9,24,870/2.4868]	3,71,911.70

Classification of lease:

Parameter 1:

The present value of lease payment i.e., Rs. 9,24,870 which equals 92.48% of the fair market value i.e., Rs. 10,00,000.

The present value of minimum lease payments substantially covers the fair value of the leased asset

Parameter 2:

The lease term (i.e. 3 years) covers the major part of the life of asset (i.e. 5 years).

Therefore, it constitutes a finance lease.

Computation of Unearned Finance Income:

Particulars	Rs.
Total lease payments (Rs. 3,71,911.70 x 3)	11,15,735
Add: Unguaranteed residual value	1,00,000
Gross investment in the lease	1,215,735
Less: Present value of lease payments and residual value i.e. Net Investment (Rs. 75,130 + Rs. 9,24,870)	(10,00,000)
Unearned finance income	**2,15,735**

8. *X Ltd. sold machinery having WDV of Rs. 300 lakhs to Y Ltd. for Rs. 400 lakhs and the same machinery was leased back by Y Ltd. to X Ltd. The lease back arrangement is operating lease.*

Give your comments in the following situations:

(i) Sale price of Rs. 400 lakhs is equal to fair value.

(ii) Fair value is Rs. 450 lakhs.

(iii) Fair value is Rs. 350 lakhs and the sale price is Rs. 250 lakhs.

(iv) Fair value is Rs. 300 lakhs and sale price is Rs. 400 lakhs.

(v) Fair value is Rs. 250 lakhs and sale price is Rs. 290 lakhs.

Solution:

Accounting Treatment:

S. No.	Particulars	Accounting Treatment
(i)	When sale price of Rs. 400 lakhs is equal to fair value	X Ltd. should immediately recognize the profit of Rs. 100 lakhs (i.e. 400 – 300) in its books.
(ii)	When fair value is Rs. 450 lakhs	Profit of Rs. 100 lakhs should be immediately recognized by X Ltd.

(iii)	When fair value of leased machinery is Rs. 350 lakhs & sales price is Rs. 250 lakhs	Then loss of Rs. 50 lakhs (300 – 250) to be immediately recognized by X Ltd. in its books provided loss is not compensated by future lease payment.
(iv)	When fair value is Rs. 300 lakhs & sales price is Rs. 400 lakhs	Then, profit of Rs. 100 lakhs is to be deferred and amortized over the lease period.
(v)	When fair value is Rs. 250 lakhs & sales price is Rs. 290 lakhs	Then the loss of Rs. 50 lakhs (300-250) to be immediately recognized by X Ltd. in its books and profit of Rs. 40 lakhs (290-250) should be amortized/ deferred over lease period.

AS 28: IMPAIRMENT OF ASSETS

Illustration 1

Ergo Industries Ltd. gives the following estimates of cash flows relating to Property, Plant and Equipment on 31-12-20X1. The discount rate is 15%.

Year	***Cash Flow (Rs. in lakhs)***
20X2	*4000*
20X3	*6000*
20X4	*6000*
20X5	*8000*
20X6	*4000*

Residual value at the end of 20X6 = *Rs. 1000 lakhs*

Property, Plant and Equipment purchased on 1-1-20XX = *Rs. 40,000 lakhs*

Useful life = *8 years*

Net selling price on 31-12-20X1 = *Rs. 20,000 lakhs*

Calculate on 31-12-20X1:

(a) Carrying amount at the end of 20X1

(b) Value in use on 31-12-20X1

(c) Recoverable amount on 31-12-20X1

(d) Impairment loss to be recognized for the year ended 31-12-20X1

(e) Revised carrying amount

(f) Depreciation charge for 20X2.

Note: *The year 20XX is the immediate preceding year before the year 20X0.*

Solution

Calculation of value in use

Year	***Cash Flow***	***Discount as per 15%***	***Discounted cash flow***
20X2	4,000	0.870	3,480
20X3	6,000	0.756	4,536
20X4	6,000	0.658	3,948

20X5	8,000	0.572	4,576
20X6	4,000	0.497	1,988
20X6	(residual) 1,000	0.497	497
			19,025

(a) **Calculation of carrying amount:**

Original cost = Rs. 40,000 lakhs

Depreciation for 3 years = [(40,000-1000)×3/8] = Rs. 14,625 lakhs

Carrying amount on 31-12-20X1 = [40,000-14,625] = Rs. 25,375 lakhs

(b) **Value in use = Rs. 19,025 lakhs**

(c) Recoverable amount = higher of value in use and net selling price i.e. Rs. 20,000 lakhs.

Recoverable amount = Rs. 20,000 lakhs

(d) **Impairment Loss** = Rs. (25,375-20,000) = Rs. 5,375 lakhs

(e) **Revised carrying amount** = Rs. (25,375-5,375) = Rs. 20,000 lakhs

(f) **Depreciation charge for 20X2** = (20,000-1000)/5 = Rs. 3,800 lakhs

Illustration 2

X Ltd. is having a plant (asset) carrying amount of which is Rs. 100 lakhs on 31.3.20X1. Its balance useful life is 5 years and residual value at the end of 5 years is Rs. 5 lakhs. Estimated future cash flow from using the plant in next 5 years are:

For the year ended on	Estimated cash flow (Rs. in lakhs)
31.3.20X2	50
31.3.20X3	30
31.3.20X4	30
31.3.20X5	20
31.3.20X6	20

Calculate "value in use" for plant if the discount rate is 10% and also calculate the recoverable amount if net selling price of plant on 31.3.20X1 is Rs. 60 lakhs.

Solution

Present value of future cash flow

Year ended	*Future Cash Flow*	*Discount @ 10% Rate*	*Discounted cash flow*
31.3.20X2	50	0.909	45.45
31.3.20X3	30	0.826	24.78
31.3.20X4	30	0.751	22.53
31.3.20X5	20	0.683	13.66
31.3.20X6	20	0.620	12.40
			118.82
Present value of residual price on 31.3.20X6 = 5 × 0.620			3.10
Present value of estimated cash flow by use of an asset and residual value, which is called "value in use".			121.92

If net selling price of plant on 31.3.20X1 is Rs. 60 lakhs, the recoverable amount will be higher of Rs. 121.92 lakhs (value in use) and Rs. 60 lakhs (net selling price), hence recoverable amount is Rs. 121.92 lakhs.

Illustration 3

G Ltd., acquired a machine on 1st April, 20X0 for Rs. 7 crore that had an estimated useful life of 7 years. The machine is depreciated on straight line basis and does not carry any residual value. On 1st April, 20X4, the carrying value of the machine was reassessed at Rs. 5.10 crore and the surplus arising out of the revaluation being credited to revaluation reserve. For the year ended March, 20X6, conditions indicating an impairment of the machine existed and the amount recoverable ascertained to be only Rs. 79 lakhs. You are required to calculate the loss on impairment of the machine and show how this loss is to be treated in the books of G Ltd. G Ltd., had followed the policy of writing down the revaluation surplus by the increased charge of depreciation resulting from the revaluation.

Solution

Statement Showing Impairment Loss

	(Rs. in crores)
Carrying amount of the machine as on 1st April, 20X0	7.00

Depreciation for 4 years i.e. 20X0-20X1 to 20X3-20X4 $\left[\frac{7\text{ crores}}{7\text{ years}} \times 4\text{ years}\right]$	(4.00)
Carrying amount as on 31.03.20X4	3.00
Add: Upward Revaluation (credited to Revaluation Reserve account)	2.10
Carrying amount of the machine as on 1st April, 20X4 (revalued)	5.10
Less: Depreciation for 2 years i.e. 20X4-20X5& 20X5-20X6 $\left[\frac{5.10\text{ crores}}{3\text{ years}} \times 2\text{ years}\right]$	(3.40)

Carrying amount as on 31.03.20X6	1.70
Less: Recoverable amount	(0.79)
Impairment loss	0.91
Less: Balance in revaluation reserve as on 31.03.20X6:	
Balance in revaluation reserve as on 31.03.20X4	2.10
Less: Enhanced depreciation met from revaluation reserve 20X4-20X5 & 20X5-20X6=[(1.70 – 1.00) x 2 years] (1.40)	
Impairment loss set off against revaluation reserve balance as per para 58 of AS 28 "Impairment of Assets"	(0.70)
Impairment Loss to be debited to profit and loss account	0.21

Illustration 4

X Ltd. purchased a Property, Plant and Equipment four years ago for ₹*. 150 lakhs and depreciates it at 10% p.a. on straight line method. At the end of the fourth year, it has revalued the asset at* ₹*. 75 lakhs and has written off the loss on revaluation to the profit and loss account. However, on the date of revaluation, the market price is* ₹*. 67.50 lakhs and expected disposal costs are* ₹*. 3 lakhs. What will be the treatment in respect of impairment loss on the basis that fair value for revaluation purpose is determined by market value and the value in use is estimated at* ₹*. 60 lakhs?*

Solution

Treatment of Impairment Loss

As per para 57 of AS 28 "Impairment of assets", if the recoverable amount (higher of net selling price and its value in use) of an asset is less than its carrying amount, the carrying amount of the asset should be reduced to its recoverable amount. In the given case, net selling price is Rs. 64.50 lakhs (Rs. 67.50 lakhs – Rs. 3 lakhs) and value in use is Rs. 60 lakhs. Therefore, recoverable amount will be Rs. 64.50 lakhs. Impairment loss will be calculated as Rs. 10.50 lakhs [Rs. 75 lakhs (Carrying Amount after revaluation - Refer Working Note) less Rs. 64.50 lakhs (Recoverable Amount)].

Thus impairment loss of Rs. 10.50 lakhs should be recognised as an expense in the Statement of Profit and Loss immediately since there was downward revaluation of asset which was already charged to Statement of Profit and Loss.

Working Note:

Calculation of carrying amount of the Property, Plant and Equipment at the end of the fourth year on revaluation

	(Rs. in lakhs)
Purchase price of a Property, Plant and Equipment	150.00
Less: Depreciation for four years [(150 lakhs / 10 years) x 4 years]	(60.00)
Carrying value at the end of fourth year	90.00
Less: Downward revaluation charged to profit and loss account	(15.00)
Revalued carrying amount	75.00

Reference: The students are advised to refer the full text of AS 28 "Impairment of Assets" (issued 2002).

Scenario based Questions

5. *A publisher owns 150 magazine titles of which 70 were purchased and 80 were self-created. The price paid for a purchased magazine title is recognized as an intangible asset. The costs of creating magazine titles and maintaining the existing titles are recognized as an expense when incurred. Cash inflows from direct sales and advertising are identifiable for each magazine title. Titles are managed by customer segments. The level of advertising income for a magazine title depends on the range of titles in the customer segment to which the magazine title relates. Management has a policy to abandon old titles before the end of their economic lives and replace them immediately with new titles for the same customer segment. What is the cash-generating unit for an individual magazine title?*

Solution:

It is likely that the recoverable amount of an individual magazine title can be assessed. Even though the level of advertising income for a title is influenced, to a certain extent, by the other titles in the customer segment, cash inflows from direct sales and advertising are identifiable for each title. In addition, although titles are managed by customer segments, decisions to abandon titles are made on an individual title basis. Therefore, it is likely that individual magazine titles generate cash inflows that are largely independent of each other and that each magazine title is a separate cash-generating unit.

6. *An asset does not meet the requirements of environment laws which have been recently enacted. The asset has to be destroyed as per the law. The asset is carried in the Balance Sheet at the year end at Rs. 6,00,000. The estimated cost of destroying the asset is Rs. 70,000. How is the asset to be accounted for?*

Solution:

As per AS 28 "Impairment of Assets", impairment loss is the amount by which the carrying amount of an asset exceeds its recoverable amount, where recoverable amount is the higher of an asset's net selling price* and its value in use•. In the given case, recoverable amount will be nil [higher of value in use (nil) and net selling price (negative Rs. 70,000)]. Thus impairment loss will be calculated as Rs. 6,00,000 [carrying amount (Rs. 6,00,000) – recoverable amount (nil)]. Therefore, asset is to be fully impaired and

impairment loss of Rs. 6,00,000 has to be recognized as an expense immediately in the statement of Profit and Loss as per para 58 of AS 28.

Further, as per para 60 of AS 28, When the amount estimated for an impairment loss is greater than the carrying amount of the asset to which it relates, an enterprise should recognise a liability if, and only if, that is required by another Accounting Standard. Hence, the entity should recognize liability for cost of disposal of Rs. 70,000 as per AS 10 & 29.

7. *Venus Ltd. has a fixed asset, which is carried in the Balance Sheet on 31.3.20X1 at Rs. 500 lakhs. As at that date the value in use is Rs. 400 lakhs and the net selling price is Rs. 375 lakhs.*

From the above data:

(i) Calculate impairment loss.

(ii) Prepare journal entries for adjustment of impairment loss.

(iii) Show, how impairment loss will be shown in the Balance Sheet.

Solution:

1. (i) Recoverable amount is higher of value in use Rs. 400 lakhs and net selling price Rs. 375 lakhs.

Recoverable amount = Rs. 400 lakhs

Impairment loss = Carried Amount – Recoverable amount

= Rs. 500 lakhs – Rs. 400 lakhs = Rs. 100 lakhs.

(ii) Journal Entries

	Particulars	**Dr. Amount (Rs. in lakhs)**	**Cr. Amount (Rs. in lakhs)**
(i)	Impairment loss account Dr. To Provision for Accumulated Impairment Loss Account (Being the entry for accounting impairment loss)	100	100

(ii)	Profit and loss account Dr. To Impairment loss (Being the entry to transfer impairment loss to profit and loss account)	100	 100

(iii) Balance Sheet of Venus Ltd. as on 31.3.20X1

	(Rs. in lakhs)
Fixed Asset	
Asset less depreciation	500
Less: Impairment loss	(100)
	400

8. *Good Drugs and Pharmaceuticals Ltd. acquired a sachet filling machine on 1st April, 20X1 for Rs. 60 lakhs. The machine was expected to have a productive life of 6 years. At the end of financial year 20X1-20X2 the carrying amount was Rs. 41 lakhs. A short circuit occurred in this financial year but luckily the machine did not get badly damaged and was still in working order at the close of the financial year. The machine was expected to fetch Rs. 36 lakhs, if sold in the market. The machine by itself is not capable of generating cash flows. However, the smallest group of assets comprising of this machine also, is capable of generating cash flows of Rs. 54 crore per annum and has a carrying amount of Rs. 3.46 crore. All such machines put together could fetch a sum of Rs. 4.44 crore if disposed. Discuss the applicability of Impairment loss.*

Solution:

As per provisions of AS 28 "Impairment of Assets", impairment loss is not to be recognized for a given asset if its cash generating unit (CGU) is not impaired. In the given question, the related cash generating unit which is group of asset to which the damaged machine belongs is not impaired; and the recoverable amount is more than the carrying amount of group of assets. Hence there is no need to provide for impairment loss on the damaged sachet filling machine.

9. *From the following details of an asset*

   (i) Find out impairment loss

   (ii) Treatment of impairment loss

   (iii) Current year depreciation

   Particulars of asset:

Cost of asset	Rs. 56 lakhs
Useful life period	10 years
Salvage value	Nil
Current carrying value	Rs. 27.30 lakhs
Useful life remaining	3 years
Recoverable amount	Rs. 12 lakhs
Upward revaluation done in last year	Rs. 14 lakhs

Solution:

According to AS 28 "Impairment of Assets", an impairment loss on a revalued asset is recognised as an expense in the statement of profit and loss. However, an impairment loss on a revalued asset is recognised directly against any revaluation surplus for the asset to the extent that the impairment loss does not exceed the amount held in the revaluation surplus for that same asset.

Impairment Loss and its treatment	Rs.
Current carrying amount (including revaluation amount of Rs. 14 lakhs)	27,30,000
Less: Current recoverable amount	(12,00,000)
Impairment Loss	15,30,000
Impairment loss charged to revaluation reserve	14,00,000
Impairment loss charged to profit and loss account	1,30,000

After the recognition of an impairment loss, the depreciation (amortization) charge for the asset should be adjusted in future periods to allocate the

asset's revised carrying amount, less its residual value (if any), on a systematic basis over its remaining useful life.

In the given case, the carrying amount of the asset will be reduced to Rs. 12,00,000 after impairment. This amount is required to be depreciated over remaining useful life of 3 years (including current year). Therefore, the depreciation for the current year will be Rs. 4,00,000.

10. *A plant was acquired 15 years ago at a cost of Rs. 5 crores. Its accumulated depreciation as at 31st March, 20X1 was Rs. 4.15 crores. Depreciation estimated for the financial year 20X1-20X2 is Rs. 25 lakhs. Estimated Net Selling Price as on 31st March, 20X1 was Rs. 30 lakhs, which is expected to decline by 20 per cent by the end of the next financial year.*

    Its value in use has been computed at Rs. 35 lakhs as on 1st April, 20X1, which is expected to decrease by 30 per cent by the end of the financial year.

    (i) Assuming that other conditions for applicability of the impairment Accounting Standard are satisfied, what should be the carrying amount of this plant as at 31st March, 20X2?

    (ii) How much will be the amount of write off for the financial year ended 31st March, 20X2?

    (iii) If the plant had been revalued ten years ago and the current revaluation reserves against this plant were to be Rs. 12 lakhs, how would you answer to questions (i) and (ii) above?

    (iv) If the value in use was zero and the enterprise were required to incur a cost of Rs. 2 lakhs to dispose of the plant, what would be your response to questions (i) and (ii) above?

Solution:

As per AS 28 "Impairment of Assets", if the recoverable amount* of an asset is less than its carrying amount, the carrying amount of the asset should be reduced to its recoverable amount and that reduction is an impairment loss. An impairment loss on a revalued asset is recognized as an expense in the statement of profit and loss. However, an impairment loss on a revalued asset is recognised directly against any revaluation surplus for the asset to the extent that the impairment loss does not exceed the amount held in the

revaluation surplus for that same asset.

In the given case, recoverable amount (higher of asset's net selling price and value in use) will be Rs. 24.5 lakhs on 31.3.20X2 according to the provisions of AS 28 [Refer working note].

		(Rs. in lakhs)
(i)	Carrying amount of plant (after impairment) as on 31st March, 20X2	24.50
(ii)	Amount of write off (impairment loss) for the financial year ended 31st March, 20X2 [Rs. 60 lakhs – Rs. 24.5 lakhs]	35.50
(iii)	If the plant had been revalued ten years ago	
	Debit to revaluation reserve	12.00
	Amount charged to profit and loss account (Rs. 35.50 lakhs – Rs. 12 lakhs)	23.50
(iv)	If Value in use is zero	
	Value in use (a)	Nil
	Net selling price (b)	(-)2.00
	Recoverable amount [higher of (a) and (b)]	Nil
	Carrying amount (closing book value)	Nil
	Amount of write off (impairment loss) (Rs. 60 lakhs – Nil)	60.00
	Entire book value of plant will be written off and charged to profit and loss account.	

Working Note:

Calculation of Closing Book Value, Estimated Net Selling Value and Estimated Value in Use of Plant at 31st March, 20X2

	(Rs. in lakhs)
Opening book value as on 1.4.20X1 (Rs. 500 lakhs – Rs. 415 lakhs)	85
Less: Depreciation for financial year 20X1–20X2	(25)

Closing book value as on 31.3.20X2	60
Estimated net selling price as on 1.4.20X1	30
Less: Estimated decrease during the year (20% of Rs. 30 lakhs)	(6)
Estimated net selling price as on 31.3.20X2	24
Estimated value in use as on 1.4.20X1	35.0
Less: Estimated decrease during the year (30% of Rs. 35 lakhs)	(10.5)

Theoretical questions and Answer

ACCOUNTING STANDARD 16 BORROWING COSTS

1. *When capitalization of borrowing cost should cease as per Accounting Standard 16? Explain the provision.*

Ans: Capitalization of borrowing costs should cease when substantially all the activities necessary to prepare the qualifying asset for its intended use or sale are complete. An asset is normally ready for its intended use or sale when its physical construction or production is complete even though routine administrative work might still continue. If minor modifications such as the decoration of a property to the user's specification, are all that are outstanding, this indicates that substantially all the activities are complete. When the construction of a qualifying asset is completed in parts and a completed part is capable of being used while construction continues for the other parts, capitalisation of borrowing costs in relation to a part should cease when substantially all the activities necessary to prepare that part for its intended use or sale are complete.

2. *H Ltd. incurs borrowing costs for the purpose of construction of a qualifying asset for its own use. The construction gets completed on May 31, 20X1. However, decoration work is under process which is expected to be completed by November 20X1 after which H Ltd. will be able to start using the said asset for its own use. H Ltd. wants to capitalize the eligible borrowing costs incurred up to November 20X1.*

Ans: The capitalization of borrowing costs shall cease when substantially all the activities necessary to prepare the qualifying assets for its intended use or sale is completed. In the given case, H Ltd. should capitalize borrowing costs only up to May 31, 20X1. The borrowing cost incurred thereafter cannot be capitalized as the asset was ready for its intended use on May 31, 20X1. The fact that decoration work was being carried out should not be considered as the asset was ready for its intended use on May 31, 20X1.

3. *ABC Ltd. is in the process of getting an entertainment park constructed. For this purpose, it has taken loan from a bank. The said park consists of several rides and facilities, each of which can be used individually. Three fourth part of the park has been constructed and can be opened up for public, while construction on the remaining part is continuing. Whether the capitalization of borrowing cost should continue for the whole park until construction continues?*

Ans: ABC Ltd. is in process of constructing an entertainment park which consists of several rides and facilities that can operate independently for their intended use. Even though the park as whole is not complete, the individual facilities are ready for their intended use.

The cessation of capitalization depends upon the nature of the qualifying assets, particularly where the qualifying assets consists of various parts. There are qualifying assets where each part is capable of being used while the construction continues on other parts. There are qualifying assets where all parts have to be completed before any earlier completed part can be put to use.

Since in the given scenario, the individual facilities are capable of operating independently and are ready for their intended use, therefore the borrowing costs shall cease to be capitalized for the three-fourth part of the project.

ACCOUNTING STANDARD 19 LEASES

1. *Explain the types of lease as per AS 19.*

For the purpose of accounting AS 19, classifies leases into two categories as follows:

1. Finance Lease

2. Operating Lease

Finance Lease:

It is a lease, which transfers substantially all the risks and rewards incidental to ownership of an asset to the lessee by the lessor but not the legal ownership.

As per para 8 of the standard, in following situations, the lease transactions are called Finance lease:

- The lessee will get the ownership of leased asset at the end of the lease term.
- The lessee has an option to buy the leased asset at the end of the lease term at price, which is lower than its expected fair value at the date on which option will be exercised.
- The lease term covers the major part of the life of asset even if title is not transferred.
- At the beginning of lease term, present value of minimum lease rental covers the initial fair value.
- The asset given on lease to lessee is of specialized nature and can only be used by the lessee without major modification.

Operating Lease:

It is lease, which does not transfer all the risks and rewards incidental to ownership.

2. *Explain the accounting treatment for a sale and leaseback transaction under Operating lease.*

As per AS 19, where sale and leaseback results in operating lease, then the accounting treatment in different situations is as follows:

Situation 1: Sale price = Fair Value

Profit or loss should be recognized immediately. Situation 2: Sale Price < Fair Value

Profit should be recognized immediately. The loss should also be recognized immediately except that, if the loss is compensated by future lease payments at below market price, it should be deferred and amortized in proportion to the lease payments over the period for which the asset is expected to be used.

Situation 3: Sale Price > Fair Value

The excess over fair value should be deferred and amortized over the period for which the asset is expected to be used.

3. *What do you understand by the term "Interest rate implicit on lease"?*

As per para 3 of AS 19 'Leases' the interest rate implicit in the lease is the discount rate that, at the inception of the lease, causes the aggregate present value of:

(a) the minimum lease payments under a finance lease from the standpoint of the lessor; and

(b) any unguaranteed residual value accruing to the lessor, to be equal to the fair value of the leased asset.

4. *What are the disclosures requirements for operating leases by the lessee as per AS-19?*

As per AS 19, lessees are required to make following disclosures for operating leases:

(a) the total of future minimum lease payments under non-cancelable operating leases for each of the following periods:

(i) not later than one year;

(ii) later than one year and not later than five years;

(iii) later than five years;

(b) the total of future minimum sublease payments expected to be received under non-cancelable subleases at the balance sheet date;

(c) lease payments recognized in the statement of profit and loss for the period, with separate amounts for minimum lease payments and contingent rents;

(d) sub-lease payments received (or receivable) recognized in the statement of profit and loss for the period;

(e) a general description of the lessee's significant leasing arrangements including, but not limited to, the following:

(i) the basis on which contingent rent payments are determined;

(ii) the existence and terms of renewal or purchase options and escalation clauses; and

(iii) restrictions imposed by lease arrangements, such as those concerning dividends, additional debt, and further leasing.

AS 26 INTANGIBLE ASSETS

1. What is meant by Intangible Assets and what are the important factors to consider the recognition of item as an Intangible asset? What is the recognition criteria in accordance with the provisions of AS 26?

Ans: An intangible asset is an identifiable non-monetary asset, without physical substance, held for use in the production or supply of goods or services, for rental to others, or for administrative purposes. Below are the 3 key ingredients to be satisfied to cover an item as an intangible asset under this standard:

- identifiability,
- control over a resource and
- expectation (i.e. probable – 50% plus) of future economic benefits flowing to the enterprise.

The recognition of an item as an intangible asset requires an enterprise to demonstrate that the item meets the definition of an intangible asset and recognition criteria set out as below:

a. It is probable that the future economic benefits that are attributable to the asset will flow to the enterprise; and

b. The cost of the asset can be measured reliably.

2. What is the measurement criteria at the time of initial recognition of Intangible assets

acquired through separate acquisition?

Ans: If an intangible asset is acquired separately, the cost of the intangible asset can usually be measured reliably. This is particularly so when the purchase consideration is in the form of cash or other monetary assets.

The cost of an intangible asset comprises:

- its purchase price,
- any import duties and other taxes (other than those subsequently recoverable by the enterprise from the taxing authorities), and
- any directly attributable expenditure on making the asset ready for its intended use. Directly attributable expenditure includes, for example, professional fees for legal services.
- Any trade discounts and rebates are deducted in arriving at the cost.

3. What is the criteria for recognition and measurement of Internally generated intangible assets. Describe which kind of cost is considered for capitalisation with respect to provisions of AS 26. Whether the same applies for internally generated goodwill also?

Ans: To assess whether an internally generated intangible asset meets the criteria for recognition, an enterprise classifies the generation of the asset into 2 phases:

- Research Phase &
- Development Phase

Research Phase - The expenses related to Research phase is expensed off in statement of Profit and loss.

Development Phase - Development is the application of research findings or other knowledge to a plan or design for the production of new or substantially improved materials, devices, products, processes, systems or services prior to the commencement of commercial production or use.

An intangible asset arising from development (or from the development phase of an internal project) should be recognised if, and only if, an enterprise can demonstrate all of the conditions given in para 6.15.

Cost of an Internally Generated Intangible Asset

The cost of an internally generated intangible asset is the sum of expenditure incurred from the time when the intangible asset first meets the recognition criteria. Reinstatement of expenditure recognised as an expense in previous annual financial statements or interim financial reports is prohibited.

The cost of an internally generated intangible asset comprises all expenditure that can be directly attributed, or allocated on a reasonable and consistent basis, to creating, producing and making the asset ready for its intended use from the time when the intangible asset first meets the recognition criteria.

Internally generated goodwill is not recognised as an asset because it is not an identifiable

resource controlled by the enterprise that can be measured reliably at cost.

ACCOUNTING STANDARD 28 IMPAIRMENT OF ASSETS

1. *Write short note on impairment of asset and its application to inventory.*

Ans: The objective of AS 28 'Impairment of Assets' is to prescribe the procedures that an enterprise applies to ensure that its assets are carried at no more than their recoverable amount. An asset is carried at more than its recoverable amount if its carrying amount exceeds the amount to be recovered through use or sale of the asset. If this is the case, the asset is described as impaired and this Standard requires the enterprise to recognize an impairment loss.

- If carrying amount < = Recoverable amount : Asset is not impaired
- If carrying amount > Recoverable amount : Asset is impaired Impairment Loss = Carrying Amount – Recoverable Amount

Recoverable amount is the higher of net selling price and its value in use

This standard should be applied in accounting for the impairment of all assets, other than (i) inventories (AS 2, Valuation of Inventories); (ii) assets arising from construction contracts (AS 7, Accounting for Construction Contracts); (iii) financial assets, including investments that are included in the scope of AS 13, Accounting for Investments; and (iv) deferred tax assets (AS 22, Accounting for Taxes on Income). AS 28 does not apply to inventories, assets arising from construction contracts, deferred tax assets or investments because other accounting standards applicable to these assets already contain specific requirements for recognizing and measuring the impairment related to these assets.